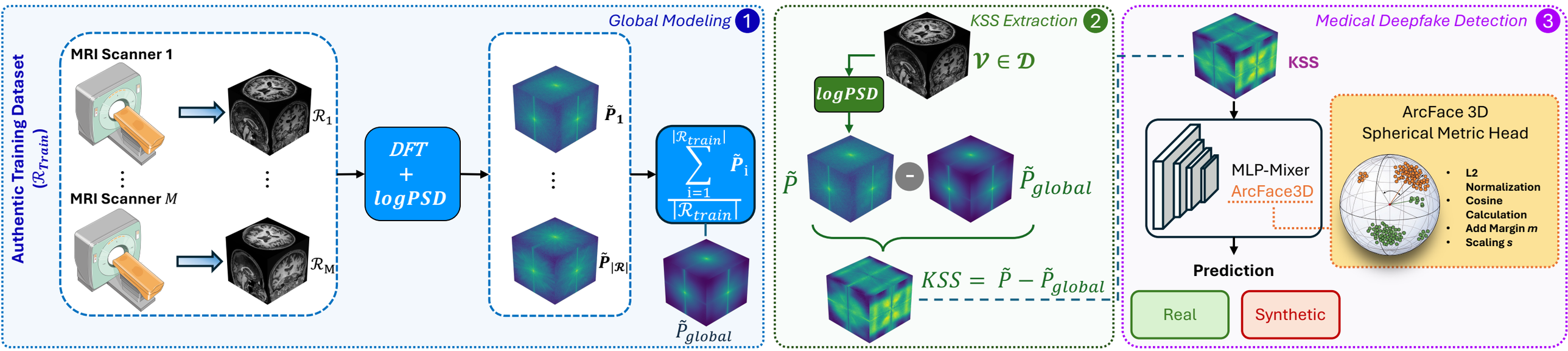

Global Modeling 1
Authentic Training Dataset ($\mathcal{R}_{Train}$)
MRI Scanner 1
MRI Scanner M
$\mathcal{R}_1$
$\mathcal{R}_M$
DFT + logPSD
$\tilde{P}_1$
$\tilde{P}_{|\mathcal{R}|}$
$\frac{\sum_{i=1}^{|\mathcal{R}_{train}|} \tilde{P}_i}{|\mathcal{R}_{train}|}$
$\tilde{P}_{global}$
KSS Extraction 2
$\mathcal{V} \in \mathcal{D}$
logPSD
$\tilde{P}$
$\tilde{P}_{global}$
$KSS = \tilde{P} - \tilde{P}_{global}$
Medical Deepfake Detection 3
KSS
MLP-Mixer
ArcFace3D
Prediction
Real
Synthetic
ArcFace 3D
Spherical Metric Head
L2 Normalization
Cosine Calculation
Add Margin m
Scaling s